

Velocity Guided Amplification of View Rotation for Seated VR Scene Exploration

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ABSTRACT

This paper presents a velocity guided amplification approach for head rotation in VR headsets, enabling the amplification factor to be dynamically changed according to head rotation velocity, and keep it within the comfort range. We first conducted experiments to investigate the effects of head rotation velocity on human sense towards the virtual view rotation and propose the velocity guided amplification approach. We then performed extensive evaluation by comparing our amplification method with existing linear mapping methods that use constant amplification factors. Results demonstrate that users achieve the best performance on given tasks with less discomfort when using our technique.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User models

1 INTRODUCTION AND RELATED WORK

When using Virtual Reality (VR) headsets while sitting, users might feel uncomfortable if they have to turn their head and body physically beyond a certain angle, as a large turning angle could cause neck strain or awkward twisting in fixed seats. Semi-natural viewing and travelling methods have been investigated for better view control to avoid that problem. Researchers have found that when visual, vestibular and proprioceptive sensory signals are conflicting in one's brain, visual modality often dominates in the estimation of self-motion. They have successfully applied this visual dominance effect in Redirected Walking (RDW) [6]. Similarly, redirecting head rotation could also produce desired perceptual illusion with small and comfortable physical head rotation. Amplifying head movements with constant [1] factors or non-linear mapping functions [5], which are termed as rotation gains [7] and dynamic rotation gains [2], have shown a significantly better performance on visual search tasks. However, a too large amplification factor (i.e. 4x as tested in [1]) has been also found to damper the task performance, as well as introduce sickness [4, 5]. Especially in non-linear mapping methods [5], a large factor could lead the amplified viewing rotation exceeding 180 degrees, because they make even larger amplification for greater head turning angles.

In this respect, further research is needed towards establishing a zone of amplification factor to modify the head movements that could offer comfortable experiences. Furthermore, according to

the research on how velocity affects users' sensitivity to the scene motion [3], the comfort zone of amplification factor may also differ with different velocities of head rotation, which have not been investigated before. In this paper, we design a technique that allows users to view the full 360-degree virtual range by rotating their heads with comfort amplification factors, which are determined by head rotation velocities.

Due to the difficulty of conducting user study during Covid-19, we chose Google Cardboard V2 with an inserted Android mobile phone as our experiment devices. Their portability makes it possible to collect data remotely. Three-degree-of-freedom (3DoF) head tracking was enabled using the IMU in Android phones. The APP for the experiment was developed using Android Studio and Google Cardboard SDK, which ran in an Android system with a resolution of 2160×1080 pixels at a refresh rate of 60Hz with 90-degree Field-of-View.

2 VELOCITY GUIDED AMPLIFICATION

We first sought to determine the highest and lowest working speeds ($^{\circ}/s$) when people are using VR devices, as well as the maximum comfortable angle of head rotation. Through our pilot study, we decided to use $26.7^{\circ}/s$, $62.2^{\circ}/s$ and their mid value $41.1^{\circ}/s$ as low/high/mid speeds for further experiments.

Based on the above three head rotation speeds, we designed a free-viewing task to study the effects of physical head rotation velocity on rotation amplification. We gradually increased or decreased the amplification factor when participants rotated their heads in each trial. Participants were asked to report whether the rotation of the virtual view matched their expectation. We recorded the lower and upper bounds of amplification factor that could provide matched view rotation, as well as the maximum factor value that still made them feel comfortable. The amplification factor started from 1.0 and we tuned it up or down by a step of 0.05. We recruited 15 participants for this experiment (10 males, 5 females, age 18-20, ϕ 19.6).

Our experiment included 3×3 tests where users are required to use the three speeds (low/mid/high) to rotate their heads under the following three different sensing conditions: the lower and upper bounds of amplification factor where virtual view rotations match participants' expectations and the maximum amplification factor where participants still feel comfortable. We define *imperceptible zone* as the range from the lower bound to the upper bound, and *comfort zone* as the range from the lower bound to the maximum amplification factor.

Table 1 shows the means of the defined amplification factors (lower bound, upper bound and largest comfortable value) and corresponding rotation speeds of virtual view under different head rotation speeds. The rotation speeds of virtual view are calculated by multiplying the head rotation speed by its amplification factor. Although amplification factor decreases along with faster head rotation, we noted that the rotation speed of virtual view still increases along with head rotation speed. The outcome verifies our hypothesis that the velocity of head rotation affects the perception of rotation speed

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Head rotation speed (degree/s)	Amplification factor			View rotation speed(degree/s)		
	Lower bound	Upper bound	Largest comfort	Lower bound	Upper bound	Largest comfort
26.7	2.02	2.46	2.95	53.93	65.42	79.03
41.1	1.81	2.23	2.55	74.39	91.65	104.81
62.2	1.61	1.90	2.22	100.14	118.18	138.08

Table 1: The means for the bounds of imperceptible zone and comfort zone of amplification factor according to different head rotation velocities.

of virtual view.

To the best of our knowledge, there is no existing method studying the comfort zone of amplification factor and in particular, how the amplification factor applies on the velocity of head rotation. Our experiments showed that $[2.02, 2.22]$ is the comfort zone for all working speeds of head rotation. A factor falling out of the comfort zone may cause sickness, especially for a factor higher than 3, which was corroborated by the work of Sargunam et al. [5].

Although people can view 360 degrees with an amplification factor of 2.02, when turning their head with a maximum angle of nearly 90 degrees in seated VR applications, the comfortable rotation angle is less than 90°. Therefore, we aimed to maintain the needed physical rotation angle within a smaller range than 90° by proposing a velocity guided amplification mechanism, in which the amplification factor is altered by a decreasing piecewise linear function. The amplification factor f_c can be written as a function of head rotation velocity v as follows:

$$f_c(v) = \begin{cases} 2.95, & v < 26.7 \\ 2.55 + \frac{41.1 - v}{41.1 - 26.7} \times (2.95 - 2.55), & 26.7 < v < 41.1 \\ 2.22 + \frac{62.2 - v}{62.2 - 41.1} \times (2.55 - 2.22), & 41.1 < v < 62.2 \\ 2.22, & v > 62.2 \end{cases} \quad (1)$$

Using the above function of amplification by velocity, people are capable of view the full 360 degrees with the maximum physical rotation angle within the range of 61° to 81°, depending on their head rotation speeds. Therefore, this approach decreases the rotation angle needed for exploring the whole scene. Note that to ensure the virtual view can return to the neutral position when people's heads are back to the physical neutral position, we use a constant amplification factor calculated by dividing the current virtual view angle by current head rotation angle when users try to rotate back. This constant amplification factor mostly falls between 2.22 and 2.95.

3 EVALUATION

We conducted an experiment to evaluate our velocity guided head rotation amplification method by comparing it with the amplification method using constant factors, which are 1x, 2x, 3x, and 4x respectively. We ask our participants to explore the virtual scene to complete two tasks: counting and searching. We measured their accuracy (counting task only) and completion time of assigned tasks, and asked them to complete a questionnaire to rate different methods in terms of rotation speed, fatigue and overall evaluation that uses a 5-Likert scale. The experiment was completed by 13 participants (11 male, 2 female, age 19-28, $\phi 21.5$). Each of them completed both tasks with all five different rotation amplification schemes. Since we focused on seated VR experiences, participants sat in a non-rotating chair, with little allowed movements of the upper body.

The counting task is devised to measure users' performance and experience when exploring a static VR scene. We generated n objects (n is a random number from 13 to 15) in three types in the virtual scene for each trial. Participants are required to count the number of objects of each type in all five conditions. The searching task

aims to assess users' performance and experience while engaging in searching and targeting activities. We asked participants to rotate their head to find the only ball in the scene, target at it and then eliminate it by pressing the button on the cardboard. When the ball disappeared, the next one would be immediately generated at another position. Every participant needed to eliminate 10 balls totally.

Quantitative assessment for different methods was conducted in our evaluation experiment. Results showed that our velocity-guided implementation provided the best performance in various ways considering both tasks. The results of the Friedman test showed a main effect of our method on speed, fatigue and overall preference for both counting and searching tasks ($p < 0.05$ for all related tests). In the counting task, our system achieved the lowest counting error rate and the least required time. In the searching task, the results of the RM-ANOVA found our implementation was significantly better in terms of task completion time ($F = 4.88, p < 0.005$).

4 CONCLUSION AND FUTURE WORK

In this paper, we introduce velocity as an amplification parameter for seated VR scenarios. We found that the factor decreases with the increment of rotation speed. Based on that, we propose our velocity-guided mapping amplification method through linear interpolation. The quantitative evaluation shows that our method outperforms other existing techniques in both counting and searching tasks, in terms of both participants' performance and level of comfort. Therefore, we believe that velocity guided amplification technique has significant potentials for seated VR scenarios and other VR application contexts.

In the future, we plan to validate the advantages of using velocities to drive the rotation amplification in commercial HMDs. We would also like to conduct more studies on imperceptible zone as well as the influence of scene complexity on people's amplification preference.

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